ENGR 111
Introduction to Engineering
Class #10

A common basis

Somebody hands you a table. The units do not match.

Nobody hands you one set of units

Real projects collect numbers from people who never agreed on units.

Battery capacity. One supplier quotes amp-hours. The next quotes watt-hours.

Torque. Pound-feet on the old drawing, newton-meters on the new one.

Fuel economy. Miles per gallon here, liters per hundred kilometers there.

Two pumps

Both move the same water. One is cheaper to buy, the other to run.

	Pump A	Pump B
Flow	40 gal/min	40 gal/min
Purchase price	$900	$1,500
Power draw	1.5 kW	1.1 kW
Assumed use	6 h/day, 300 days/year	6 h/day, 300 days/year
Average electricity	$0.14/kWh	$0.14/kWh

Need a cost-to-own column to compare.

Choose the basis first

The basis is what everything gets converted into. Here, dollars per year.

$$1.5\ \mathrm{kW} \times 6\ \mathrm{h/day} \times 300\ \mathrm{day/yr} \times \$0.14/\mathrm{kWh} = \$378/\mathrm{yr}$$

$$1.1\ \mathrm{kW} \times 6\ \mathrm{h/day} \times 300\ \mathrm{day/yr} \times \$0.14/\mathrm{kWh} = \$277/\mathrm{yr}$$

$$\mathrm{kW} \times \mathrm{h/day} \times \mathrm{day/yr} \times \$/\mathrm{kWh} \rightarrow \$/\mathrm{yr}$$

The answer

$600 more to buy. $101 a year cheaper to run.

	Pump A	Pump B
Purchase price	$900	$1,500
Energy per year	2,700 kWh	1,980 kWh
Energy cost per year	$378	$277

$600 ÷ $101 per year ≈ 6 years

More than six years in service: buy B. Less: buy A.

Nobody in the table knows how long it runs.

What our simple table did not tell us

Maintenance, spare parts, downtime. None of it is in the table.

Halve the running hours and the payback doubles. About 12 years.

How can you get more accurate numbers for the table?

The price tag is not the price

The smallest number on a shelf can be the most expensive thing on it.

	As sold	Per gallon
Gasoline	\$4.10 / gal	\$4.10
Milk	\$4.99 / gal	\$4.99
Bottled water	\$2.00 / 16.9 oz bottle	**\$15.15**

128 fluid ounces in a gallon. That one conversion reverses the order.